

LOCALIZATION IN THE COMMUTANT, STAGE 12:

the network as its own two-tier system.

Self-localization and the internal Eldan clock, a hyperbolic upper tier, the gate gas, and the memory as a cumulant series stratified by border codimension

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Abstract

Stages 9–11 studied the bias-free He-initialised RELU network with Gaussian input inside a two-tier framework imposed from outside. The lower tier is the network’s gates; the upper tier is Eldan’s stochastic localization of the input. Here we show that the network carries the same two-tier structure *internally*, and that the external framework is the internal one shifted in time. (i) *Self-localization*. Each layer of a fresh network acts as a localization step. Through every row of W_{l+1} , the law of h_l looks like a Gaussian ball whose standardized centre is $N(0, t_l)$ -distributed across rows, exactly as Eldan’s localized input looks to a first layer at time t_l . The internal clock is $t_l = |\mathbb{E}h_l|^2 / \text{Tr Cov } h_l = \cos \theta_l / (1 - \cos \theta_l)$, where θ_l is the folding angle of independent inputs. It obeys a closed recursion, and in the Fatou chart of stage 11 it advances by exactly one unit per layer. Depth and localization time are therefore one coordinate, Szekeres’ regular iteration defines fractional depth, and the stage-11 collapse theorem is this statement seen from outside. Measured on a width-1024 network, the field law, gate variance, clock and log-norm variance match the predictions. (ii) *Cooling*. A localized state is the Gibbs state of $H = |x|^2/2$ at $\beta = 1 + t$, so the network anneals itself on the schedule $T_l = 1 - \cos \theta_l \approx \theta_l^2/2$. At finite width the temperature splits into an angular part and a radial part, $T_l \approx (1 - \cos \theta_l) + V_l/4$, where V_l is the across-input variance of $\log |h_l|^2$. This variance saturates at $44.6/n$ because its increments decay like θ^2 . The radial part takes over at Fatou time $\tau_* \approx 2\sqrt{n}$. (iii) *A hyperbolic upper tier*. Under the Fisher–Rao metric, the per-neuron Gaussian laws form a hyperbolic plane. Positive homogeneity is translation along the kink geodesic $\{\mu = 0\}$, and rectification is a skew product over the signed distance to it. Criticality means zero drift along this geodesic, and the collective (norm) mode is the finite-width diffusion along it. Measured, this diffusion, not the birth of cumulants at the walls, is what makes deep pre-activations non-Gaussian at width 1024: each neuron’s skewness is linear in its field ($R^2 \approx 0.9$) and its excess kurtosis is flat, both proportional to V with one efficiency ≈ 0.7 at every depth. (iv) *The gate gas*. The lower tier is a dichotomized-Gaussian lattice gas. Its Ising couplings are $c_{ab}\varphi(r_a)\varphi(r_b)/(p_aq_ap_bq_b)$, with a Mattis component whose stored pattern is the field itself. Every wall-localized one-body quantity obeys a *wall-density law*: it equals $(\int Q) \sin(\theta/2)/\sqrt{\pi}$, so it decays harmonically in Fatou time and its total over depth grows like $\log \tau$. (v) *Tier coupling*. At every layer an exact Euler split writes the mean as an upper-tier term plus a gate–field covariance. The non-Gaussian corrections form an Edgeworth series. We prove that each Edgeworth term is a sum over strata of the wall arrangement, and that the joint cumulant entering a stratum is that of exactly the fields whose walls meet there: the Kikuchi level equals the codimension. Normal crossings carry nothing. Two-body memory lives only on the bent crossings, where stage 11 found the non-abelian isotropy $\mathfrak{aff}(1)$. (vi) *Modular structure*. The layer’s Gaussian state is a KMS state on the fibre relation of h_l . Its extremal decomposition is the law of h_l , its conditional expectation onto the centre is the posterior, and the centres form a decreasing tower. The forward state localizes

while the backward (sensitivity) state delocalizes with the power law $(\tau_l/\tau_L)^3$. Every claim is labelled as proved, proved at infinite width, conjectural or an analogy. The identities are checked by `scripts/verify_stage12.py`.

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1 The realisation in one page

1. One layer is a two-tier system (Section 2).

- The upper tier is the quasi-free projection (μ_l, C_l) of the pre-activations.

- The lower tier is the gate gas.
- The up map is the gating $h = Dz$ followed by W_{l+1} , and the closure is the second-order moment chain.
- The memory is the non-quasi-free part of the law.
- An exact Euler split, $\mathbb{E}f = \langle \mathbb{E}J_l, \mu_l \rangle + \sum_a \text{Cov}(J_{l,a}, z_{l,a})$, separates the tiers at every layer. At the input the whole mean is the lower-tier border functional of stage 11.

2. The network localizes itself (Section 3). Through every row of the next layer, the law of h_l is a Gaussian ball whose standardized centre is $N(0, t_l)$ across rows, with $t_l = |m_l|^2 / \text{Tr } \Sigma_l \rightarrow \cos \theta_l / (1 - \cos \theta_l)$. This is exactly what a first layer sees of Eldan’s localized input at time t_l , with rows playing the role of localization paths. The clock obeys $t' = \mathbb{E}m_1^2 / \mathbb{E}s_1^2$, which is the folding map in disguise, and in the Fatou chart it advances by one per layer. Depth is localization time: Szekeres’ regular iteration gives fractional depth, and external localization to time t is fractional depth $\Phi_F(\arccos \frac{t}{1+t}) - \Phi_F(\pi/2)$. Localized states are Gibbs states of $|x|^2/2$ at $\beta = 1 + t$, so the network anneals itself on $T_l = 1 - \cos \theta_l \approx 9\pi^2/2\tau_l^2$. At finite width a radial temperature $V_l/4$ (the log-normal norm) is added. It explains the measured clock at $l = 16$ (0.0779 against $0.0705 + 0.0080$) and takes over at $\tau_* \approx 2\sqrt{n}$.

3. The upper tier is hyperbolic (Section 4). With the Fisher–Rao metric the per-neuron laws form a hyperbolic plane. The field is the signed distance to the kink geodesic $\{\mu = 0\}$, $r = \sqrt{2} \sinh(d/\sqrt{2})$. Positive homogeneity is translation along that geodesic, and rectification is a skew product with base map $R = m_1/s_1 > 0$, which pushes every neuron to the positive side. Criticality means zero drift along the geodesic. The finite-width diffusion along it has across-input increments $\frac{4}{n}(\frac{3}{2} - \frac{J_2(\theta)}{2\pi}) \approx 4\theta^2/n$, which are summable, so $nV_l \rightarrow 44.6$ rather than growing like $5l$. The collective mode (trace share $\approx tV/4$) is this diffusion. It compresses the field law and skews the pre-activations: skewness is linear in the field ($R^2 \approx 0.9$) and the excess kurtosis is flat, both proportional to V .

4. The lower tier is a gate gas with a wall-density law (Section 5). The gas is a dichotomized Gaussian with Ising couplings $c_{ab}\varphi_a\varphi_b/(p_aq_ap_bq_b)$, and its Mattis component stores the field as its pattern. Every wall-localized one-body quantity equals $(\int Q) \sin(\theta_l/2)/\sqrt{\pi} \approx 2.66(\int Q)/\tau_l$. The lower tier is therefore active harmonically in Fatou time, and its accumulated activity grows like $\log \tau$. Hermite cancellation suppresses neuron averages of the marginal memory, but not its mean square.

5. The memory is a cumulant series stratified by codimension (Section 6). Each Edgeworth term is a sum over chain strata of the wall arrangement, and the cumulant on a stratum is the joint cumulant of the fields whose walls meet there. One-body cumulants live on walls and two-body cumulants on bent crossings: the Kikuchi level is the codimension. Normal crossings carry nothing, so the Gaussian closure is exact for them, and pair memory sits exactly at stage 11’s $\text{aff}(1)$ points. The third-order formula is checked to one standard error, and the strata cancel. Third cumulants are born at rate $0.718/\tau$.

6. Modular structure (Section 7). The Gaussian state is KMS on the fibre relation of h_l . Its extremal decomposition is the law of the representation, the conditional expectation onto the centre is the posterior, and the centres $L^\infty(h_l)$ form a decreasing tower. Coarse-graining the centre and cooling are interchangeable on the next layer. The forward state localizes while the backward state delocalizes, as $\prod(1 - \theta_{k-1}/\pi) \approx (\tau_l/\tau_L)^3$; this is confirmed on the width-1024 network near the output and is about 20% faster near the input.

2 Two tiers inside one layer

Notation. As in stage 11: $W_l \in \mathbb{R}^{n \times n}$ with i.i.d. $N(0, 2/n)$ entries, $h_0 = x \sim \gamma = N(0, I_n)$, $z_l = W_l h_{l-1}$, $h_l = \text{relu}(z_l)$, gates $g_l = \mathbf{1}\{z_l > 0\}$, $D_l = \text{diag}(g_l)$. For a fixed network, $m_l = \mathbb{E}h_l$, $\Sigma_l = \text{Cov } h_l$, $\mu_l = \mathbb{E}z_l = W_l m_{l-1}$ and $C_l = \text{Cov } z_l = W_l \Sigma_{l-1} W_l^\top$; expectations are over x unless stated. Neuron a of layer l has *scale* $s_a = C_{l,aa}^{1/2}$, *field* $r_a = \mu_{l,a}/s_a$ and gate probability $p_a = \Pr(g_{l,a} = 1)$, with $q_a = 1 - p_a$. The folding angles are $\theta_0 = \pi/2$ and $\theta_l = F(\theta_{l-1})$, where $\cos F(\theta) = (\sin \theta + (\pi - \theta) \cos \theta)/\pi$ (the arc-cosine kernel). Φ_F is the Fatou coordinate of stage 11 ($\Phi_F \circ F = \Phi_F + 1$, $\Phi_F(\theta) = 3\pi/\theta + \frac{3}{2} \log \theta + O(\theta)$, $\Phi_F(\pi/2) = 7.7548$). The standard normal density and distribution function are φ and Φ . Single-neuron moments of $\text{relu}(r + \xi)$, $\xi \sim N(0, 1)$:

$$m_1(r) = \varphi(r) + r\Phi(r), \quad m_2(r) = (1 + r^2)\Phi(r) + r\varphi(r), \quad s_1(r)^2 = m_2(r) - m_1(r)^2.$$

2.1 The abstract two-tier system

Definition 2.1 (two-tier system). A *two-tier system* consists of the following.

- An *upper tier*: a manifold $\mathcal{U}$ of quasi-free (Gaussian) states.
- A *lower tier*: a finite lattice gas $\{0, 1\}^n$ fibred over $\mathcal{U}$.
- A *down map* $\mathcal{D} : \mathcal{U} \rightarrow \mathcal{P}(\{0, 1\}^n)$, sending a state to the conditional law of the gas.
- An *up map* $\mathcal{R}$, sending the joint law of gas and field to the next upper-tier state.

The *closure* is the composite $\mathcal{R} \circ \mathcal{D}$ in which the gas is integrated against the quasi-free state. The *memory* is the difference between the true next state and the closure.

In stages 9 and 10 the lower tier was the network and the upper tier was Eldan's localization path of the input. The tiers were coupled from outside, through conditioning on $\mathcal{F}_t$. The point of this note is that one layer of the network is already a two-tier system of this kind.

2.2 The cycle of one layer

- **Upper tier at layer l :** the quasi-free projection (μ_l, C_l) of the law of z_l . By positive homogeneity only rays matter, and each neuron contributes a point $(\mu_{l,a}, s_a)$ of the upper half plane. The upper tier is a cloud of n points in a hyperbolic plane (Section 4).
- **Lower tier at layer l :** the gates $g_l \in \{0, 1\}^n$. Under the quasi-free law their joint law is the dichotomized Gaussian $g_a = \mathbf{1}\{r_a + \xi_a > 0\}$, with $\xi \sim N(0, c)$ and $c_{ab} = C_{l,ab}/(s_a s_b)$ (Section 5).
- **Up map:** $h_l = D_l z_l$, the field gated by the gas. Then (m_l, Σ_l) is pushed forward, $(\mu_{l+1}, C_{l+1}) = (W_{l+1} m_l, W_{l+1} \Sigma_l W_{l+1}^\top)$.

- **Closure:** compute (m_l, Σ_l) from (μ_l, C_l) as if z_l were Gaussian. Then $m_{l,a} = s_a m_1(r_a)$, $\Sigma_{l,aa} = s_a^2 s_1(r_a)^2$, and $\Sigma_{l,ab}$ is the bivariate rectified covariance. This is the second-order moment chain of stages 9 and 10.
- **Memory:** the non-quasi-free part of the law of z_l . Its size is measured by the negentropy $D(\text{law}(z_l) \parallel N(\mu_l, C_l))$, and its effect on means is given by the cumulants of order ≥ 3 (Section 6).

2.3 The exact split

Proposition 2.2 (Euler split; proved). *Let $f = c^\top h_L$, or any output that is positively homogeneous of degree one in z_l ; every deeper output is. Put $J_l = \partial f / \partial z_l$, a piecewise constant row vector. Then $f = \langle J_l, z_l \rangle$ pointwise, and*

$$\mathbb{E}f = \underbrace{\langle \mathbb{E}J_l, \mu_l \rangle}_{\text{upper tier}} + \underbrace{\sum_a \text{Cov}(J_{l,a}, z_{l,a})}_{\text{lower tier}}.$$

If z_l is Gaussian, Stein's identity turns the lower-tier term into $\langle C_l, \mathbb{E}\nabla_{z_l}^2 f \rangle$. The Hessian of the piecewise linear f is a measure on the walls of layers $\geq l$, so the lower-tier term is then a border functional. At the last layer the split is, neuron by neuron,

$$\mathbb{E}h_{L,a} = p_a \mu_a + \text{Cov}(g_a, z_a), \quad \text{and under the closure} \quad s_a m_1(r_a) = \underbrace{s_a r_a \Phi(r_a)}_{\text{upper}} + \underbrace{s_a \varphi(r_a)}_{\text{lower}}.$$

At $l = 0$, where $z_0 := x$ and $\mu_0 = 0$, the upper term vanishes: $\mathbb{E}f = \mathbb{E}\langle x, \nabla f \rangle = \mathbb{E}\Delta f$. The whole mean is then a lower-tier border functional, which is stage 11's border formula.

Proof. Euler's identity for degree-one homogeneous piecewise linear maps gives $f = \langle J_l, z_l \rangle$. Take expectations and write $\mathbb{E}[J_a z_a] = \mathbb{E}J_a \mu_a + \text{Cov}(J_a, z_a)$. For Gaussian z , Stein gives $\text{Cov}(J_a, z_a) = \sum_b C_{ab} \mathbb{E}\partial_b J_a$, and $\partial_b J_a = \partial_a \partial_b f$. The last-layer form uses $J_{L,a} = c_a g_a$ and $\text{Cov}(\mathbf{1}\{z > 0\}, z) = s^2 p_z(0) = s\varphi(r)$ for $z \sim N(rs, s^2)$. $\square$

The same split holds with the post-activations and the Putnam–Treviño pairing of stage 11. Since $f = \langle h_l, \delta_l \rangle$ with $\delta_l = \partial f / \partial h_l$, we have $\mathbb{E}f = \langle m_l, \mathbb{E}\delta_l \rangle + \sum_i \text{Cov}(h_{l,i}, \delta_{l,i})$. The upper-tier term pairs the two harmonic states' means, and the lower-tier term their fluctuations. Section 7 shows that the forward state localizes with depth, while the backward state delocalizes towards the input.

3 The internal Eldan clock

3.1 Eldan's localization seen through one row

Eldan's localization of γ tilts it by $\exp(\langle y, x \rangle - t|x|^2/2)$. The result is the ball $\gamma_{t,y} = N(y/(1+t), I/(1+t))$, and along the localization path ($y_t = tX + B_t$) the centre $y_t/(1+t)$ has law $N(0, \frac{t}{1+t}I)$. Project onto a unit row w : the ball becomes $N(\nu, \frac{1}{1+t})$, with standardized centre $\nu\sqrt{1+t} \sim N(0, t)$. Two independent samples from the same localized ball have overlap $t/(1+t) + o(1)$ in high dimension. So a first layer that sees the input at localization time t sees, through every row, a unit ball at a field distributed as $N(0, t)$.

Lemma 3.1 (field law; proved). *For $t \geq 0$, $\mathbb{E}_{r \sim N(0,t)} \Phi(r)\Phi(-r) = \frac{\arccos \frac{t}{1+t}}{2\pi}$.*

Proof. $\Phi(r)\Phi(-r) = \Pr(\xi_1 < r, \xi_2 < -r)$ for independent standard ξ_i . Averaging over r , the pair $(\xi_1 - r, \xi_2 + r)$ is centred Gaussian with variances $1 + t$ and correlation $-t/(1 + t)$. The orthant formula gives $\frac{1}{4} - \frac{1}{2\pi} \arcsin \frac{t}{1+t} = \frac{1}{2\pi} \arccos \frac{t}{1+t}$. $\square$

3.2 The internal clock

Definition 3.2. For a fixed network, the *internal clock* and *internal overlap* of layer l are

$$t_l = \frac{|m_l|^2}{\text{Tr } \Sigma_l}, \quad q_l = \frac{|m_l|^2}{\mathbb{E}|h_l|^2} = \frac{t_l}{1 + t_l}.$$

Since $|m_l|^2 = \mathbb{E}\langle h_l(x), h_l(x') \rangle$ for independent inputs x, x' , q_l is the *replica overlap* of two independent inputs at depth l .

Theorem 3.3 (self-localization; proved at infinite width). *Fix l and let $n \rightarrow \infty$.*

1. $q_l \rightarrow \cos \theta_l$ and $t_l \rightarrow t_l^\infty := \cos \theta_l / (1 - \cos \theta_l)$. Equivalently $\arccos \frac{t_l}{1+t_l} = \theta_l$: the internal clock is the Eldan time whose input overlap equals the folding angle.
2. The empirical law over $a \in [n]$ of the fields $r_{l+1,a}$ of the next layer converges to $N(0, t_l^\infty)$. For almost every row, the law of $(z_{l+1,a} - \mu_{l+1,a})/s_a$ over x converges to $N(0, 1)$.
3. The mean gate variance of layer $l + 1$ converges to $\frac{1}{n} \sum_a p_a q_a \rightarrow \mathbb{E}_{N(0, t_l)} \Phi(r)\Phi(-r) = \theta_l/2\pi$.
4. The clock obeys the closed recursion

$$t_{l+1}^\infty = \frac{\mathbb{E} m_1(r)^2}{\mathbb{E} s_1(r)^2}, \quad r \sim N(0, t_l^\infty),$$

and in the chart $t \mapsto \arccos \frac{t}{1+t}$ this recursion is the folding map F .

So layer $l + 1$ looks at the law of h_l , through each of its rows, exactly as a first layer looks at Eldan's localized input at time t_l . The rows of W_{l+1} play the role of localization paths, and self-averaging over neurons replaces averaging over paths.

Proof. (1) For independent x, x' the pair $(h_l(x), h_l(x'))$ obeys the arc-cosine (NNGP) recursion. The input angle concentrates at $\pi/2$, so the normalized overlap is $\cos \theta_l$ in the limit. Also $|m_l|^2 = \mathbb{E}\langle h_l(x), h_l(x') \rangle$ because x, x' are independent given W . (2) W_{l+1} is independent of (m_l, Σ_l) , so $\mu_{l+1,a} = \langle w_a, m_l \rangle \sim N(0, 2|m_l|^2/n)$ exactly. The quadratic form $s_a^2 = w_a^\top \Sigma_l w_a$ concentrates at $\frac{2}{n} \text{Tr } \Sigma_l$ because $\|\Sigma_l\|_{op} = o(\text{Tr } \Sigma_l)$ at fixed l : the largest (collective) eigenvalue carries a share $O(1/n)$, see Proposition 4.3. The ratio therefore converges in law to $N(0, |m_l|^2/\text{Tr } \Sigma_l)$. Gaussianity of the projected law is Sudakov's (Diaconis–Freedman's) theorem for random projections of a thin-shell vector, and thin shell holds by the Hanin–Nica log-normal law of $|h_l|^2$ (Proposition 4.2). (3) follows from (2) with $p_a = \Phi(r_a)$ and Lemma 3.1. (4) Under (2), $m_{l+1,a} = s m_1(r_a)$ and $\Sigma_{l+1,aa} = s^2 s_1(r_a)^2$, which gives the ratio. For the chart, mixing over r turns $z = s(r + \xi)$ into $N(0, s^2(1 + t))$, so $s^2 \mathbb{E} m_2 = s^2(1 + t)/2$. Likewise $s^2 \mathbb{E} m_1(r)^2 = \mathbb{E}[\text{relu}(z) \text{relu}(z')]$ for two draws sharing the centre, with correlation $t/(1 + t) = \cos \theta$. By Cho–Saul this is $\frac{s^2(1+t)}{2\pi}(\sin \theta + (\pi - \theta) \cos \theta)$. Hence $\frac{t'}{1+t'} = \frac{\mathbb{E} m_1^2}{\mathbb{E} m_2} = \cos F(\theta)$. $\square$

Remark 3.4 (what is and is not proved). Items (1), (3) and (4) are the standard infinite-width (NNGP) statements, rephrased. The new content is the identification in (2): the rows of the next layer sample Eldan centres at time t_l . The projection CLT used in (2) is rigorous given thin shell. We have not written out rates, and at fixed n the collective mode makes s_a depend on μ_a (Proposition 4.3).

3.3 Depth is localization time

Corollary 3.5 (one clock; proved at infinite width). *Put $\Theta(t) = \arccos \frac{t}{1+t}$ and $\mathcal{T}(t) = \Phi_F(\Theta(t))$. Then $\mathcal{T}(t_l^\infty) = \Phi_F(\pi/2) + l$ exactly. Externally localizing the input to time t and then running l layers gives internal time t' with $\mathcal{T}(t') = \mathcal{T}(t) + l$. This is the stage-11 collapse theorem seen from inside: the gates of layer $l+1$ at external time t sit at the stage-11 Fatou time $\tau(l+1, t) = \mathcal{T}(t) + l$.*

Write $\tau_l := \Phi_F(\theta_l) = 7.755 + l$ for the Fatou time of the representation h_l ; it is also the Fatou time of the gates of layer $l+1$. Szekeres' regular iteration $F^s = \Phi_F^{-1}(\Phi_F + s)$ is the unique C^1 flow embedding of the parabolic germ that is regular at 0. It defines *fractional depth*, and Corollary 3.5 says that Eldan localization to time t is fractional depth

$$s(t) = \Phi_F(\Theta(t)) - \Phi_F(\pi/2) \sim 3\pi\sqrt{(1+t)/2} \quad (t \rightarrow \infty).$$

Exactly, $\sqrt{1+t_l} = 1/(\sqrt{2}\sin(\theta_l/2))$. One layer is worth $\Delta\sqrt{1+t} \rightarrow \sqrt{2}/3\pi = 0.150$ (stage 11), and $t_l^\infty = 0.47, 0.98, 2.13, 5.04, 8.72, 13.18$ at $l = 1, 2, 4, 8, 12, 16$. The infinite-width network is therefore the time-one map of a flow on the clock line, and the external upper tier of stages 9–10 is the same flow started earlier.

3.4 The network cools itself

Proposition 3.6 (cooling; proved). $\gamma_{t,y} \propto \exp(-(1+t)H + \langle y, x \rangle)$ with $H(x) = |x|^2/2$. So the localized ball is the Gibbs (KMS) state of H at inverse temperature $\beta = 1+t$ with external field y , and $\hbar = 1/\beta$ is a temperature. At infinite width layer l is at internal temperature

$$T_l = \frac{1}{1+t_l} = 1 - \cos \theta_l = 2 \sin^2 \frac{\theta_l}{2} \approx \frac{\theta_l^2}{2},$$

and its unresolved gate fraction $\theta_l/\pi = \frac{2}{\pi} \arcsin \sqrt{T_l/2}$ freezes like $\sqrt{T_l}$. The schedule $T_l \approx \frac{9\pi^2}{2\tau_l^2}$ is an annealing schedule in Fatou time, and it is critical: the accumulated unresolved fraction $\sum_l \theta_l/\pi \approx 3 \log \tau$ diverges logarithmically.

Proposition 3.7 (angular and radial temperature; heuristic, consistent with data). Write $h_l = \rho u$ with $\rho^2 = |h_l|^2/\mathbb{E}|h_l|^2$, and let $V_l = \text{Var}_x \log |h_l|^2$. Suppose ρ and u are independent and $\log \rho^2$ is Gaussian; independence is exact for the input's own radius, since $h_l(x) = |x| h_l(x/|x|)$. Then

$$T_l = 1 - e^{-V_l/4} \cos \angle_l = \underbrace{(1 - \cos \angle_l)}_{T_l^{\text{ang}}} + \underbrace{V_l/4}_{T_l^{\text{rad}}} + O(V_l T_l^{\text{ang}}),$$

where $\cos \angle_l$ is the overlap of the normalized representations, which tends to $\cos \theta_l$. The radial temperature is a finite-width effect. It is the log-normal norm, and it does not vanish with depth: by Proposition 4.2, V_l saturates at $44.6/n$. The two temperatures cross at $\theta^2/2 = V_\infty/4$, i.e. at Fatou time $\tau_* \approx 3\pi\sqrt{2n/44.6} = 2.0\sqrt{n}$. Beyond it the clock saturates at $t \approx 4/V$, while the angle keeps shrinking (compare the stage-11 conjecture on the end of the parabolic regime).

Derivation. $q_l = |\mathbb{E}[\rho u]|^2/\mathbb{E}|h_l|^2 = (\mathbb{E}\rho)^2 \cos \angle_l$ under independence. With $\log \rho^2 \sim N(-V/2, V)$, $(\mathbb{E}\rho)^2 = e^{-V/4}$. $\square$

At $n = 1024$ and $l = 16$ the measured temperature is $T = 1/(1 + 11.84) = 0.0779$. Proposition 3.7 with the measured $V_{16} = 0.0321$ predicts $1 - e^{-0.0080} \cos \theta_{16} = 0.0779$, so the radial

part accounts for the whole gap between the measured clock 11.84 and the infinite-width 13.18. At $l = 4, 8, 12$ the same comparison is off by $+0.024, +0.010, -0.003$. That scatter has the size of the quenched fluctuations of the clock. A width scan over $l = 2, \dots, 8$ ($n = 256, 1024, 4096$; 6, 3, 2 networks) gave mean deviations $\cos \theta_l - q_l = +0.012, -0.007, -0.001$, which change sign and shrink with n : they are quenched $O(n^{-1/2})$ fluctuations of a random clock, not a bias. The radial part is a systematic $O(1/n)$ effect. At these widths it is visible only in the deepest layers, where it has grown to $V/4 \approx 0.008$.

3.5 Measured self-localization

A single He network ($n = 1024$, $L = 16$, 2×10^5 standard inputs, seed 5) gives the following; full output in the Verification section.

l	q_l meas./ $\cos \theta_l$	t_l meas./ t_l^∞	gate var. meas./ $\theta_{l-1}/2\pi$	V_l meas./pred.
1	0.3183 / 0.3183	0.47 / 0.47	0.2500 / 0.2500	0.0069 / 0.0068
2	0.4857 / 0.4937	0.94 / 0.98	0.1994 / 0.1984	0.0107 / 0.0107
4	0.6545 / 0.6810	1.89 / 2.13	0.1517 / 0.1466	0.0172 / 0.0164
8	0.8196 / 0.8344	4.54 / 5.04	0.1009 / 0.1000	0.0249 / 0.0233
12	0.8933 / 0.8971	8.38 / 8.72	0.0765 / 0.0769	0.0292 / 0.0274
16	0.9221 / 0.9295	11.84 / 13.18	0.0614 / 0.0629	0.0321 / 0.0300

The field law of the next layer is checked in Section 4. The fraction of unresolved neurons ($|r| < 1$) at $l = 4, 8, 12, 16$ is listed in Table 1.

4 The upper tier is hyperbolic

4.1 The Fisher–Rao plane and the kink geodesic

The Fisher–Rao metric on the univariate normal family $N(\mu, \sigma^2)$ is $ds^2 = (d\mu^2 + 2d\sigma^2)/\sigma^2$. In the coordinates $(\mu/\sqrt{2}, \sigma)$ it is twice the Poincaré metric, so the family is a hyperbolic plane of curvature $-\frac{1}{2}$ (Skovgaard). We call the vertical line $K = \{\mu = 0\}$, the centred balls, the *kink geodesic*: these are the laws that sit exactly on a RELU wall.

Proposition 4.1 (proved). *1. The dilations $(\mu, \sigma) \mapsto (\lambda\mu, \lambda\sigma)$ are the hyperbolic translations along K . Their orbits are the equidistant curves $\{r = \mu/\sigma = \text{const}\}$.*

2. The distance of $N(\mu, \sigma^2)$ to K is $d = \sqrt{2} \operatorname{asinh}(|r|/\sqrt{2})$. So the field is a signed hyperbolic distance, $r = \pm\sqrt{2} \sinh(d/\sqrt{2})$.

3. The single-neuron rectification $\varrho : (\mu, \sigma) \mapsto (\mathbb{E} \operatorname{relu}, \operatorname{sd} \operatorname{relu}) = \sigma(m_1(r), s_1(r))$ commutes with the translations along K . It is a skew product over the leaf space of equidistants: the base map is $R(r) = m_1(r)/s_1(r)$, and the translation cocycle is $\log s_1(r)$, i.e. the log-scale moves along K by $\log s_1(r)$.

4. $R(0) = 0.6833$, $R(r) = r + o(1)$ as $r \rightarrow +\infty$, and $R(r) \sim \sqrt{\varphi(r)/2|r|} \rightarrow 0^+$ as $r \rightarrow -\infty$. Hence $R(\mathbb{R}) \subset (0, \infty)$: after rectification every neuron lies on the positive side of K .

Proof. (1) and (2): translate to the Poincaré half-plane, where the distance from (u, σ) to the imaginary axis is $\operatorname{asinh}(|u|/\sigma)$. (3): relu is positively homogeneous. (4): use the Mills-ratio asymptotics $m_1 \sim \varphi/r^2$ and $m_2 \sim 2\varphi/|r|^3$ as $r \rightarrow -\infty$. $\square$

Item (4) is the geometric reason the clock runs forward: rectification moves the whole line of fields to one side of the kink geodesic. The next layer's mixing then spreads the cloud again, but with a positive bias in the mean direction. The aggregate effect is the ratio of averages $t' = \mathbb{E}m_1^2/\mathbb{E}s_1^2$, not the average of R^2 .

4.2 The upper tier as a cloud in the hyperbolic plane

At layer l the upper tier is the cloud $\{(\mu_{l,a}, s_a)\}_{a \leq n}$. By Theorem 3.3 its signed distances to K satisfy $\sqrt{2} \sinh(d_a/\sqrt{2}) \sim N(0, t_{l-1})$, so a typical neuron is at distance $d \approx \sqrt{2} \log \sqrt{2t} \approx \sqrt{2} \log l + O(1)$. The cloud escapes from the kink geodesic logarithmically in depth. Only its second moment in the r -chart, the clock, enters the infinite-width dynamics, and the dilation coordinate along K drops out by homogeneity.

Proposition 4.2 (criticality and diffusion along K ; drift proved, diffusion as in Hanin–Nica). *The He variance $2/n$ is the unique scale for which $\mathbb{E}|h_l|^2$ is constant in l : the barycentric log-scale has zero drift along K . At finite width, the input-dependent log-norm performs a random walk along K . For a fixed network its across-input increments have variance*

$$\text{Var}_x \left(\log \frac{|h_l|^2}{|h_{l-1}|^2} \right) = \frac{4}{n} \left(\frac{3}{2} - \frac{J_2(\theta_{l-1})}{2\pi} \right), \quad J_2(\theta) = 3 \sin \theta \cos \theta + (\pi - \theta)(1 + 2 \cos^2 \theta),$$

with $V_0 = 2/n$. Since $\frac{3}{2} - \frac{J_2(\theta)}{2\pi} = \theta^2 + O(\theta^4)$, the increments are summable in the parabolic regime and

$$nV_{16} = 30.76, \quad nV_\infty = 44.60,$$

against $2 + 5l = 82$ at $l = 16$ for the joint (weights and input) variance of Hanin–Nica. The input-dependent part of the norm is made in the first layers: deep layers act on all inputs alike, because two inputs at angle θ disagree on only a fraction θ/π of the gates.

Derivation. Per layer the norm ratio is

$$|h_l|^2/|h_{l-1}|^2 \approx \frac{2}{n} \sum_a \text{relu}(\zeta_a)^2, \quad \zeta_a \text{ standard},$$

with variance $\frac{4}{n}(\mathbb{E} \text{relu}^4 - \frac{1}{4}) = 5/n$. The across-input variance is this minus the covariance between two independent inputs, whose ζ 's are at angle θ_{l-1} . That covariance is

$$\frac{4}{n}(\mathbb{E}[\text{relu}(\zeta)^2 \text{relu}(\zeta')^2] - \frac{1}{4}) = \frac{4}{n} \left(\frac{J_2(\theta)}{2\pi} - \frac{1}{4} \right)$$

by the degree-two arc-cosine kernel. Expanding J_2 at 0 gives $3\pi - 2\pi\theta^2 + O(\theta^4)$. $\square$

Proposition 4.3 (the collective mode; heuristic, measured). *The covariance Σ_l has a collective direction along m_l , carrying the share $\approx t_l V_l/4$ of the trace. This is the norm fluctuation projected on the mean: $\langle \hat{n}, h \rangle \approx |m|(1 + \delta)$ with $\text{Var } \delta \approx V/4$, i.e. the diffusion along K . Through the next layer it makes the scale depend on the mean, $s_a^2 \approx s_{\text{bulk}}^2 + \epsilon \mu_a^2$ with $\epsilon \approx V/4$. The field law is therefore compressed,*

$$r_a = \frac{\varrho_a}{\sqrt{1 + \epsilon \varrho_a^2}}, \quad \varrho_a \sim N(0, t_{\text{bulk}}), \quad |r_a| < \epsilon^{-1/2} \approx 2/\sqrt{V}.$$

In the hyperbolic picture the cloud cannot go further than $\sqrt{2} \text{asinh } \sqrt{2/V}$ from the kink geodesic. The collective share becomes $O(1)$ when $t_l V_l/4 \sim 1$, which is again $\tau_ \approx 2\sqrt{n}$ (Proposition 3.7).*

l	Var r	t_{l-1} meas.	compressed	t_{l-1}^∞	$\Pr(r < 1)$	coll./ $tV/4$	Frob.
4	1.38	1.43	1.41	1.53	0.615 / 0.581	0.008 / 0.005	0.034
8	3.79	3.74	3.57	4.24	0.362 / 0.373	0.026 / 0.022	0.122
12	7.07	7.51	6.83	7.73	0.286 / 0.281	0.058 / 0.053	0.316
16	10.14	10.93	9.52	11.99	0.226 / 0.227	0.093 / 0.086	0.469

Table 1: Field law of layer l ($n = 1024$, one network). $\Pr(|r| < 1)$ is measured against $\text{erf}(1/\sqrt{2t_{l-1}^\infty})$, and “coll.” is the collective trace share against $tV/4$. “Compressed” is $\mathbb{E}[\varrho^2/(1 + \epsilon\varrho^2)]$ with t_{bulk} and ϵ taken from the measured clock and collective share. The last column (“Frob.”) is stage 9’s metric, the Frobenius share of the top eigenvalue of the off-diagonal covariance.

Table 1 supports the picture. The unresolved fraction $\Pr(|r| < 1)$ matches the infinite-width $\text{erf}(1/\sqrt{2t})$ to 0.001 at the deepest layers, because compression acts on large $|r|$. The measured collective trace share tracks $tV/4$ (0.093 against 0.086 at $l = 16$). At $l = 12$ and 16 the field variance lies below the measured clock, as compression predicts, but the one-parameter model over-predicts the effect: at $l = 16$ it gives 9.52, against 10.14 measured and 10.93 uncompressed. At $l = 8$ the variance exceeds the measured clock, which is within the quenched scatter. The over-prediction disappears once only a fraction $\eta \approx 0.71$ of V is counted as a common scale: Proposition 6.6 measures this efficiency independently from the skewness, and it then gives 7.11 and 10.09 against 7.07 and 10.14. The stage-9 statement that “37–48% of the off-diagonal covariance energy is collective” is the Frobenius share in the last column. It reaches 0.47 at $l = 16$ although the collective *trace* share is only 0.09: a rank-one mode of modest trace dominates the off-diagonal Frobenius norm, because the bulk off-diagonal entries are random and incoherent.

5 The lower tier: the gate gas and the wall-density law

5.1 The gate gas

For a fixed network the *gate gas* of layer l is the law of $g_l(x) \in \{0, 1\}^n$, $x \sim \gamma$. Under the quasi-free law of z_l it is the dichotomized Gaussian of Emrich–Piedmonte and Macke et al.: $g_a = \mathbf{1}\{r_a + \xi_a > 0\}$ with $\xi \sim N(0, c)$. The one-body marginals are $p_a = \Phi(r_a)$. The two-body marginals are orthant probabilities $\Phi_2(r_a, r_b; c_{ab})$.

Proposition 5.1 (Ising couplings; proved to first order in c). *The pair coupling of the gate gas (the log odds ratio of the 2×2 table) is*

$$J_{ab} = \log \frac{P_{11}P_{00}}{P_{10}P_{01}} = \frac{c_{ab} \varphi(r_a) \varphi(r_b)}{p_a q_a p_b q_b} + O(c_{ab}^2).$$

Proof. $\partial_c \Phi_2(r_a, r_b; c)|_{c=0} = \varphi(r_a) \varphi(r_b)$. A covariance ϵ in a table with margins p_a, p_b changes the log odds ratio by $\epsilon(\frac{1}{p_a p_b} + \frac{1}{p_a q_b} + \frac{1}{q_a p_b} + \frac{1}{q_a q_b}) = \epsilon/(p_a q_a p_b q_b)$. $\square$

The correlations c_{ab} have two parts.

- A *bulk* part of size $O(n^{-1/2})$ with random signs, which is Sherrington–Kirkpatrick-like.
- A *collective* part $c_{ab}^{\text{coll}} \approx \epsilon r_a r_b$ coming from the norm mode (Proposition 4.3).

By Proposition 5.1 the collective part is a Mattis coupling, $J_{ab}^{\text{coll}} = \epsilon \xi_a \xi_b$, with stored pattern

$$\xi_a = \frac{r_a \varphi(r_a)}{p_a q_a} \approx r_a |r_a| \quad (|r_a| \gg 1).$$

The pattern the gate gas stores is the field itself, that is, the representation’s own mean. Its Mattis order parameter is not spontaneous. It is the input’s radial coordinate $\delta(x)$, a latent variable shared by all neurons. Given δ , the gates are conditionally independent up to the bulk couplings, which is a de Finetti form. Resolved neurons ($|r| \gg 1$) have large couplings per unit correlation but tiny variance $p_a q_a$; the gas is thermally active only near its “Fermi level” $r = 0$.

Remark 5.2 (two statistics; analogy). The field z is a quasi-free boson field (a Gaussian, i.e. a quasi-free state on a CCR algebra). The gates are fermionic occupation numbers ($g^2 = g$). In the independent-gate approximation their state is the product (gauge-invariant quasi-free) state with diagonal one-particle density $\text{diag}(\Phi(r_a))$, and the occupation is a probit-smeared Fermi function of the “energy” $-r_a$. The RELU coupling $h = g z$ is a boson–fermion (Yukawa-type) vertex with the occupation slaved to the field. The Euler split (Proposition 2.2) is its Wick decomposition, $\mathbb{E}[gz] = \mathbb{E}g \mathbb{E}z + \text{Cov}(g, z)$, and the closure is the corresponding mean-field (Hartree–Fock–Bogoliubov) decoupling. Depth lowers the density of states at the Fermi level as $1/\tau$ (Theorem 5.3).

5.2 The wall-density law

Theorem 5.3 (wall-density law; proved). *Let Q be integrable with $\int r^2 |Q| < \infty$, and $\Theta = \arccos \frac{t}{1+t}$. Then*

$$\mathbb{E}_{r \sim N(0,t)} Q(r) = \left(\int Q \right) \frac{\sin(\Theta/2)}{\sqrt{\pi}} \left(1 - \frac{m_2(Q) - 1}{2t} + O(t^{-2}) \right), \quad m_2(Q) = \frac{\int r^2 Q}{\int Q},$$

and $\sin(\Theta/2)/\sqrt{\pi} = 1/\sqrt{2\pi(1+t)}$ exactly. At layer $l + 1$, where $\Theta = \theta_l$, every wall-localized one-body quantity therefore equals

$$\left(\int Q \right) \frac{\sin(\theta_l/2)}{\sqrt{\pi}} \approx \left(\int Q \right) \frac{\theta_l}{2\sqrt{\pi}} \approx \frac{3\sqrt{\pi}}{2} \left(\int Q \right) \frac{1}{\pi} = 2.659 \left(\int Q \right) \frac{1}{\pi}.$$

Proof. Expand $e^{-r^2/2t}/\sqrt{2\pi t} = (2\pi t)^{-1/2}(1 - r^2/2t + O(t^{-2}r^4))$ and $(2\pi(1+t))^{-1/2} = (2\pi t)^{-1/2}(1 - 1/2t + O(t^{-2}))$. Use $1 + t = 1/(2\sin^2(\Theta/2))$. $\square$

quantity Q	$\int Q$	law at layer $l + 1$	check
wall density $\varphi(r)$	1	$\sin(\theta_l/2)/\sqrt{\pi}$ (exact)	$t = 13$: 0.10662181 both
gate variance $\Phi(r)\Phi(-r)$	$1/\sqrt{\pi}$	$\theta_l/2\pi + O(\theta^3)$	$t = 100$: 0.022415 vs 0.022396
third-cumulant source $k_3(r)^2$	0.2700	$0.2700 \sin(\theta_l/2)/\sqrt{\pi}$	Section 6

Corollary 5.4 (the harmonic budget; proved at infinite width). *The activity of the lower tier decays harmonically in Fatou time, and its total over depth grows like $2.659(\int Q) \log \tau$. Each e -fold of Fatou time costs the same: the lower tier, and with it the memory it gives birth to, is spread log-uniformly over depth, not concentrated in the deepest layers. For the gate variance the total over 16 layers is $\sum_{l \leq 16} \theta_{l-1}/\pi = 3.658$.*

Proposition 5.5 (Hermite cancellation; proved). $\mathbb{E}_{r \sim N(0,t)}[\text{He}_k(r)\varphi(r)] = 0$ for odd k , and for $k = 2j$

$$\mathbb{E}_{r \sim N(0,t)}[\text{He}_{2j}(r)\varphi(r)] = \frac{(-1)^j(2j-1)!!}{\sqrt{2\pi}} (1+t)^{-j-1/2}.$$

Proof. $\varphi(r)\varphi_t(r) = (2\pi(1+t))^{-1/2}\varphi_{\sigma^2}(r)$ with $\sigma^2 = t/(1+t)$. For $Y \sim N(0, \sigma^2)$, $\mathbb{E}\text{He}_{2j}(Y) = (\sigma^2 - 1)^j(2j-1)!!$, from the generating function $\mathbb{E}e^{sY-s^2/2} = e^{s^2(\sigma^2-1)/2}$. $\square$

The marginal memory of a neuron, i.e. the error of the one-neuron closure, is by the Gram-Charlier form of the Edgeworth series (Section 6)

$$\frac{\mathbb{E}\text{relu}(z_a) - s_a m_1(r_a)}{s_a} = \left(-\frac{\gamma_{1,a}}{6}\text{He}_1(r_a) + \frac{\gamma_{2,a}}{24}\text{He}_2(r_a) + \frac{\gamma_{1,a}^2}{72}\text{He}_4(r_a) \right) \varphi(r_a) + \dots,$$

where γ_1, γ_2 are the standardized skewness and excess kurtosis of z_a . Every term is wall-localized. By Proposition 5.5, its *average* over neurons is suppressed by extra powers of $T = 1/(1+t)$ when the γ 's are uncorrelated with r . Its *mean square*, the quantity an MSE score sees, obeys the wall-density law with $\int (\text{He}_k \varphi)^2$. The marginal memory is therefore a sign-alternating wall-localized field over neurons. It nearly cancels in neuron averages, but not in the MSE.

On the width-1024 network the measured marginal skewness and kurtosis of the pre-activations grow with depth: mean $|\gamma_1| = 0.043, 0.070, 0.095$, and mean $|\gamma_2| = 0.041, 0.055, 0.066$, at $l = 8, 12, 16$. The sampling noise is about 0.004 and 0.009. Their source is the collective mode, not the birth law (Proposition 6.6). With the empirical cumulants, the three Gram-Charlier wall terms reproduce the empirical one-neuron closure error neuron by neuron: the correlation is 0.983, 0.993 and 0.996 at $l = 8, 12, 16$, and the residual is $\leq 1.7 \times 10^{-4}$ in every $|r|$ -bin. The error vanishes for $|r| \geq 3$ (rms 4×10^{-5}) and peaks at $1 \leq |r| < 2$ (rms 1.9×10^{-3} at $l = 16$). Both sides are computed from the same 2×10^5 samples, so this checks the expansion and the wall-localized profile. It does not fix the absolute size of the true memory, which is comparable to the Monte Carlo floor ($1.4\text{--}1.6 \times 10^{-3}$).

6 Coupling the tiers: the memory as a stratified cumulant series

6.1 The Edgeworth series of a suffix

Fix a layer l and let $S : \mathbb{R}^n \rightarrow \mathbb{R}$ be the suffix from z_l to the output; it is piecewise linear and positively homogeneous. Let κ_k be the cumulant tensors of z_l and $\mathbb{E}_G$ the expectation under the quasi-free projection $N(\mu_l, C_l)$. The density of z_l is formally $\exp(\sum_{k \geq 3} (-1)^k \kappa_k \cdot \partial^k / k!)$ applied to the Gaussian density. Integrating by parts gives the formal Edgeworth series (McCullagh)

$$\mathbb{E}f = \mathbb{E}_G S + \frac{1}{6} \langle \kappa_3, \mathbb{E}_G \nabla^3 S \rangle + \frac{1}{24} \langle \kappa_4, \mathbb{E}_G \nabla^4 S \rangle + \frac{1}{72} \langle \kappa_3^{\otimes 2}, \mathbb{E}_G \nabla^6 S \rangle + \dots$$

The first term is the closure (upper tier composed with lower tier). The rest is the memory. Equivalently, by the cumulant expansion $\mathbb{E}[z_a G(z)] = \sum_k \frac{1}{k!} \kappa_{k+1}(a, \cdot) \mathbb{E} \nabla^k G$ (Lytova-Pastur), the lower-tier term of the Euler split is $\sum_a \text{Cov}(J_a, z_a) = \sum_{k \geq 1} \frac{1}{k!} \langle \kappa_{k+1}, \mathbb{E} \nabla^{k+1} f \rangle$. Its $k = 1$ term is the Stein (Gaussian) part of Proposition 2.2.

For piecewise linear S each $\nabla^k S$, $k \geq 2$, is a distribution. Every derivative that hits a gate produces a δ (or a derivative of δ) on that neuron's wall. So each term of $\nabla^k S$ is supported on an intersection of walls, a stratum of the wall arrangement, whose codimension is the number of distinct δ factors.

6.2 Which strata carry memory

Proposition 6.1 (chain support; proved). *Every stratum that carries a term of $\nabla^k S$ is a chain: a set of walls in which every wall except the deepest lies in the input cone of a deeper wall of the set. Strata spanned by walls of a single layer (normal crossings) carry no term. In particular, if the suffix is one layer with a linear readout, $S = \sum_b W_b \text{relu}(z_b)$, then all mixed derivatives vanish and the memory is purely marginal: $\mathbb{E}f - \mathbb{E}_G S = \sum_b W_b (\mathbb{E} \text{relu}(z_b) - \mathbb{E}_G \text{relu}(z_b))$.*

Proof. A gate of neuron a depends on z_b only through the input cone of a . Differentiating $\sigma'(y_a) \prod(\cdots)$ in z_b and z_c for two neurons of one layer produces a δ on z_b or z_c only together with a derivative of a deeper gate that both feed. So the two shallow walls never appear without a deeper wall in their common output cone. Induct over layers. $\square$

Theorem 6.2 (stratified third order; proved). *Consider a two-layer suffix $S(z) = \sigma(y)$ with $y = \sum_b W_b \sigma(z_b)$, with σ smooth and $g = W \odot \sigma'(z)$ the pulled-back deep normal,*

$$\langle \kappa, \nabla^3 S \rangle = \underbrace{\sigma'''(y) \kappa(g, g, g)}_{\text{deep wall}} + 3 \underbrace{\sigma''(y) \sum_c W_c \sigma''(z_c) \kappa(g, e_c, e_c)}_{\text{bent crossings}} + \underbrace{\sigma'(y) \sum_c W_c \sigma'''(z_c) \kappa_{ccc}}_{\text{hub layer}}$$

for every symmetric 3-tensor κ . In the RELU limit ($\sigma' = \mathbf{1}_{>0}$, $\sigma'' = \delta$, $\sigma''' = \delta'$) the three terms live on:

- the deep wall $\{y = 0\}$ (codimension 1), where they pair with the third cumulant of the linearized deep field $\langle g, z \rangle$;
- the bent crossings $\{y = 0\} \cap \{z_c = 0\}$ (codimension 2), where they pair with the two-body cumulant $\kappa(\langle g, z \rangle, z_c, z_c)$ and are weighted by the bending $T_c = W_c$ of stage 11;
- the shallow walls $\{z_c = 0\}$ (codimension 1), where they pair with the one-body cumulants $\kappa_3(z_c)$ and are weighted by the deep gate.

Proof. Chain rule: $\partial_a S = \sigma'(y) g_a$; $\partial_a \partial_b S = \sigma''(y) g_a g_b + \delta_{ab} \sigma'(y) W_a \sigma''(z_a)$; one more derivative gives the three groups, and the middle one appears in three index positions. Contract with symmetric κ . $\square$

softplus β	Hermite IBP	deep wall	bent crossings	hub layer	sum	difference (s.e.)
5	0.1242	-0.0159	-0.1514	0.2896	0.1224	0.0018 (0.0028)
20	0.1677	0.2421	-0.5730	0.5007	0.1698	-0.0022 (0.0039)

The check uses a 3-neuron Gaussian z with a random symmetric κ and 2×10^7 samples. The left side is $\mathbb{E}[S(z) \text{He}_\kappa(z)]$ by Hermite integration by parts; the right side is the three strata. They agree within one standard error. The strata cancel substantially, so no single one dominates. A truncation that keeps one stratum and drops the others is not justified by this structure.

Corollary 6.3 (Kikuchi level = codimension; proved at third order, by Proposition 6.1 in general). *The cumulant that enters a stratum is the joint cumulant of exactly the fields whose walls meet there: deep and shallow neurons along a chain, with the deep field linearized. One-body cumulants live on codimension-one walls, two-body cumulants on codimension-two bent crossings, and j -body cumulants on codimension- j chains. The cluster (Kikuchi) level of the memory equals the codimension of its border. The Gaussian closure is exact on normal-crossing arrangements. Two-body memory lives only where stage 11 found the non-abelian isotropy $\mathfrak{aff}(1)$.*

This places the stage-9 decomposition: its hub-diagonal pairing is the hub layer, its coherent (mean-field) memory is the deep-wall term, and the bent-crossing term is new.

6.3 Birth of the third cumulant

Proposition 6.4 (birth law; proved at infinite width). *For $z \sim N(rs, s^2)$, $\kappa_3(\text{relu } z) = s^3 k_3(r)$ with $k_3 = m_3 - 3m_1 m_2 + 2m_1^3$, $m_3(r) = (r^3 + 3r)\Phi(r) + (r^2 + 2)\varphi(r)$, and $k_3(0) = c_3 = (\pi + 2)/(2\pi)^{3/2} = 0.32646$. The mean-square source of one-body third cumulants at layer $l + 1$ is*

$$B_l = \mathbb{E}_{N(0, t_l)} k_3(r)^2 = 0.2700 \frac{\sin(\theta_l/2)}{\sqrt{\pi}} (1 + O(\theta_l^2)) \approx \frac{0.718}{\tau_l}.$$

Measured against $0.2700/\sqrt{2\pi t}$: 0.0802/0.1077, 0.0493/0.0539, 0.0290/0.0299, 0.0151/0.0152 at $t = 1, 4, 13, 50$. By the harmonic budget the total one-body third-cumulant source over depth grows like $0.718 \log \tau$, at the same rate for every e -fold of Fatou time.

Remark 6.5 (which route the measured skewness takes). Transported through W_{l+1} with random signs, the one-body source alone gives a standardized skewness of order n^{-1} . That is far below the measured 0.095 at $l = 16$. The measured skewness must therefore come from many-body cumulants, and the natural candidate is the collective mode. If $h = (1 + \delta)\tilde{h}$ with $\text{Var } \delta = v \approx V/4$ independent of the bulk, then $z_a - \mu_a = \delta\mu_a + s\xi_a(1 + \delta)$, which gives

$$\kappa_3 = 6\mu_a s^2 v, \quad \kappa_4 = 12s^4 v + O(v^2), \quad \gamma_1 \approx 6v r_a \approx \frac{3}{2} V r_a, \quad \gamma_2 \approx 12v \approx 3V.$$

So skewness should be linear in the field, and excess kurtosis flat in it, both proportional to V .

Proposition 6.6 (collective-mode cumulant law; measured). *On the width-1024 network the marginal cumulants of the pre-activations follow this law in form. At $l = 8, 12, 16$ the regression of γ_1 on r over the 1024 neurons has slopes 0.0252, 0.0302, 0.0336, with $R^2 = 0.86, 0.90, 0.91$ and intercepts below 10^{-4} . The predicted slopes $\frac{3}{2}V_{l-1}$ are 0.0350, 0.0421, 0.0474, a constant ratio $\eta = 0.72, 0.72, 0.71$. The excess kurtosis is flat in r ($R^2 \leq 0.11$ for a fit in r^2), at levels 0.040, 0.053, 0.059 against $3V_{l-1} = 0.070, 0.084, 0.095$. The same efficiency η , inserted in the compression of the field law (Proposition 4.3), gives $\text{Var } r = 3.64, 7.11, 10.09$ against the measured 3.79, 7.07, 10.14. This removes the over-prediction in Table 1 at $l = 12, 16$. A fraction $\eta \approx 0.7$ of the across-input log-norm variance acts as a common random scale on the next layer's pre-activations.*

The law identifies the dominant finite-width source of marginal non-Gaussianity at $L = 16$, $n = 1024$. It is not the one-body birth at the walls. It is the radial diffusion along the kink geodesic, acting as a common random scale on the bulk noise. Because RELU commutes with that scale, this part of the memory is a one-dimensional mixture rather than a genuine many-body effect (Section 8, item 6).

7 Modular structure: KMS states, centres and the two clocks

7.1 The layer state as a KMS state on the fibre relation

Let $\mathcal{R}_l = \{(x, x') : h_l(x) = h_l(x')\}$ be the fibre relation of layer l . Its fibres are polyhedral; for $l = 1$ and invertible W_1 the fibre through x is an orthant of dimension equal to the number of closed gates. Equip it with Lebesgue measure on the fibres (the Haar system) and the real cocycle $c(x, x') = H(x) - H(x')$, $H = |x|^2/2$. The cocycle generates the one-parameter group $\sigma_s(k)(x, x') = e^{is c(x, x')} k(x, x')$ on kernels k on $\mathcal{R}_l$.

- Proposition 7.1** (KMS states and the centre; proved in the measured setting). *1. The state ω_μ of a measure μ , given by $\omega_\mu(k) = \int k(x, x) d\mu(x)$, is (σ, β) -KMS iff the conditional measures of μ on the fibres are $\propto e^{-\beta H} dx$ (Renault’s correspondence, Feldman–Moore in the measured case). So the KMS_β states are parametrized by probability measures on the quotient $h_l(\mathbb{R}^n)$, and the extremal ones are the fibre Gibbs states ω_h , one for each value h of the representation.*
- 2. The Gaussian state ω_γ is KMS_1 , and its extremal decomposition is the law of the representation: $\omega_\gamma = \int \omega_h d\mu_l(h)$ with $\mu_l = (h_l)_* \gamma$.*
- 3. The centre of the von Neumann algebra of $\mathcal{R}_l$ is $\mathcal{Z}_l = L^\infty(h_l)$, the functions of the representation. The ω_γ -preserving conditional expectation onto it is the posterior $E_l = \mathbb{E}[\cdot | h_l]$.*
- 4. Eldan’s localized states $\gamma_{t,y}$ are KMS at $\beta = 1 + t$ for the perturbed cocycle $c - \frac{1}{\beta} \langle y, x - x' \rangle$. At infinite width layer l is at $\beta_l = 1 + t_l$ (Proposition 3.6).*

Proof. (1) The KMS condition for σ on the groupoid algebra says that the Radon–Nikodym cocycle of μ along $\mathcal{R}_l$, relative to the Haar system, is $e^{-\beta c}$. That is the stated property of the conditionals, and extremality is ergodicity, i.e. a single fibre. (2) is disintegration of γ along h_l . (3) $\mathcal{R}_l$ has a proper (smooth) quotient, so its von Neumann algebra is the direct integral of $B(L^2(\text{fibre}))$ over $h_l(\mathbb{R}^n)$, and the centre is the diagonal. (4) is Proposition 3.6. $\square$

Since h_{l+1} is a function of h_l , $\mathcal{R}_l \subset \mathcal{R}_{l+1}$. The algebras increase, and the centres form a decreasing tower

$$L^\infty(\gamma) \supset \mathcal{Z}_1 \supset \mathcal{Z}_2 \supset \cdots \supset \mathcal{Z}_L \ni f, \quad E_{l+1} E_l = E_{l+1}.$$

The mean is $\omega_\gamma(f)$ for f in the smallest centre. This is “localization in the commutant” (stage 9) realized inside the network: each layer is a central decomposition, and the representation at depth l is the spectrum of its centre. The forward pass computes, for one input, the extremal KMS component it belongs to.

Remark 7.2 (what the modular picture adds). The upper tier of stages 9–10 (the localized balls $\gamma_{t,y}$) and the network’s own tower (the centres $\mathcal{Z}_l$) are two families of states for one modular Hamiltonian H . The external family varies β at fixed algebra. The internal family varies the algebra at fixed $\beta = 1$, and by Theorem 3.3 its effect on the next layer equals a change of β to $1 + t_l$. Cooling (raising β) and coarse-graining (shrinking the centre) are thus interchangeable on the next layer, to leading order at infinite width. This is the operator-algebraic form of “depth is localization time”.

7.2 Two harmonic states, two clocks

Stage 11 identified the forward activations and the backward sensitivities $\delta_l = \partial f / \partial h_l$ as the two harmonic states of the network’s Bratteli diagram, with $\langle h_l, \delta_l \rangle = f$ at every layer. Their clocks run in opposite directions.

Proposition 7.3 (backward clock; infinite width, gradient independence). *For $f = c^\top h_L$, with readout c independent of the weights,*

$$\frac{|\mathbb{E} \delta_l|^2}{\mathbb{E} |\delta_l|^2} \longrightarrow \prod_{k=l+1}^L \left(1 - \frac{\theta_{k-1}}{\pi} \right) \approx \left(\frac{\tau_l}{\tau_L} \right)^3.$$

The forward state localizes with depth ($q_l = \cos \theta_l \uparrow 1$), while the backward state delocalizes as it travels towards the input, with a parabolic power law of exponent $3 = 1/(\pi a)$, where $F(\theta) = \theta - a\theta^2 + \dots$ and $a = 1/3\pi$.

Derivation. $\mathbb{E}_W \langle \delta_{k-1}(x), \delta_{k-1}(x') \rangle = 2 \Pr(g_k = g'_k = 1) \langle \delta_k, \delta'_k \rangle = (1 - \theta_{k-1}/\pi) \langle \delta_k, \delta'_k \rangle$, while $\mathbb{E}|\delta_{k-1}|^2 = \mathbb{E}|\delta_k|^2$ at criticality. This uses the gradient-independence ansatz, justified by Yang's tensor programs for a readout independent of the weights. Then $\theta_{k-1}/\pi = 3/\tau + O(\tau^{-2} \log \tau)$, so the product is $\exp(-3 \sum 1/\tau) \approx (\tau_l/\tau_L)^3$. $\square$

On the width-1024 network the measured backward clock is 0.0108, 0.0354, 0.0800, 0.2489, 0.5571, 0.8798 at $l = 0, 2, 4, 8, 12, 15$. The infinite-width products are 0.0133, 0.0442, 0.0941, 0.2663, 0.5589, 0.8743. Agreement is close near the output, and the measured state delocalizes about 20% faster near the input, where finite-width errors compound. Far out, the power law is accurate: the product over $l = 1000..10^4$ is 0.00104, against $(\Phi_F(\theta_{1000})/\Phi_F(\theta_{10^4}))^3 = 0.00102$.

The Euler split at layer l pairs the two states: $\mathbb{E}f = \langle m_l, \mathbb{E}\delta_l \rangle + \sum_i \text{Cov}(h_{l,i}, \delta_{l,i})$. The upper-tier (mean–mean) term needs both states localized. Near the output the forward state is localized and the backward state is trivially so. Near the input the backward state has delocalized to 1% of its energy, so the mean there is almost entirely lower tier, which is stage 11's border formula at $l = 0$. In Montanari's correspondence between stochastic localization and diffusion models, the forward (localizing) clock and the backward (delocalizing) clock are the two time directions of one process (an analogy only).

8 What the realisation unlocks

8.1 The realisation

The two-tier framework of stages 9–11 is not imposed on the network. The network realizes it, layer by layer.

	external (stages 9–10)	internal (this note)
upper tier	Eldan balls $\gamma_{t,y}$	quasi-free projection (μ_l, C_l) ; a cloud in the Fisher–Rao plane
clock	localization time t	$t_l = m_l ^2 / \text{Tr } \Sigma_l = \cos \theta_l / (1 - \cos \theta_l)$
paths	Brownian localization paths	rows of W_{l+1} (self-averaging)
temperature	$1/(1+t)$	$1 - \cos \theta_l$ (+ radial $V_l/4$ at finite width)
lower tier	gates conditioned on $\mathcal{F}_t$	the gate gas: dichotomized Gaussian, Mattis pattern = field
coupling	conditioning	Euler split; Edgeworth series stratified by codimension
algebra	conditional expectation on $\mathcal{F}_t$	centres $\mathcal{Z}_l = L^\infty(h_l)$ of the fibre relations; KMS decomposition
renormalization time	Fatou time $\tau(l, t)$	$\tau_l = \Phi_F(\theta_l) = 7.755 + l$; fractional depth

Corollary 3.5 makes the dictionary exact at infinite width: the external upper tier is the internal one started at fractional depth $s(t)$.

8.2 What it unlocks

1. **One flow for all depths.** The infinite-width network is the time-one map of Szekeres' flow on the clock line, and localization is fractional depth. A quantity calibrated at one (l, t) transfers to every (l', t') with the same Fatou time. Stage 11 stated this as an open problem ("chains in Fatou time"). At infinite width it is now a theorem for every quantity determined by the pair angle. Departures from it are finite-width signals by definition.
2. **A temperature budget.** $T_l = T_l^{\text{ang}} + T_l^{\text{rad}}$. The angular part is the parabolic, infinite-width cooling $\approx 9\pi^2/2\tau^2$. The radial part $V_l/4$ is the log-normal norm, which saturates at $44.6/(4n)$. For the competition networks ($n = 1024$, $L = 16$), $\tau_{16} = 23.8$ is far below $\tau_* \approx 2\sqrt{n} = 64$, so they are angular-dominated. Yet at $l = 16$ the radial part (0.008) is already 11% of the angular one (0.0705), and it accounts for the gap between the measured and the infinite-width clock.
3. **A harmonic budget for the lower tier.** The layer average of each wall-localized one-body quantity Q is $(\int Q) \sin(\theta_l/2)/\sqrt{\pi}$, which is about $2.66(\int Q)/\tau_l$. The memory is born at rate $0.718/\tau$ and accumulates like $\log \tau$. A correction scheme must therefore treat every layer, with weight $\propto 1/\tau_l$. Truncating to the last few layers is not justified.
4. **The memory is a sum over bent strata.** The Edgeworth corrections decompose over chain strata of the wall arrangement, with Kikuchi level equal to codimension. Normal crossings carry nothing, and pair memory lives only on bent crossings (the $\text{aff}(1)$ points). This is an algorithmic skeleton: one-body corrections on single walls (hub and deep-wall terms, both already present in stage 9), plus a pair correction on each (deep, shallow) bent crossing. Because the strata cancel, all three must be kept together.
5. **Hermite cancellation.** The neuron-averaged marginal memory is suppressed by powers of T_l , but its mean square is not. Averaged diagnostics (for example the mean of the output vector) understate the memory that an MSE score sees.
6. **The collective mode is a symmetry direction.** The norm mode is translation along the kink geodesic, and RELU commutes with it. For the input's own radius this gives an exact factorization, $\mathbb{E}h_L = \mathbb{E}|x| \cdot \mathbb{E}_{\hat{x} \sim \text{unif}} h_L(\hat{x})$. Any memory carried by the collective mode is a mixture over this translation. It can therefore be integrated out in one dimension rather than expanded in cumulants, to the extent that the radial factor decouples from the angular one.
7. **Two clocks.** The forward state localizes and the backward state delocalizes as $(\tau_l/\tau_L)^3$. The Euler split shows where the upper-tier (mean-mean) term is reliable and where the mean is a pure border functional.

8.3 Predictions

1. For a fixed He network and $l \ll \sqrt{n}$, the empirical field law of layer $l + 1$ is $N(0, t_l)$ with $t_l \rightarrow \cos \theta_l / (1 - \cos \theta_l)$. The unresolved fraction is $\Pr(|r| < 1) = \text{erf}(1/\sqrt{2t_l})$, which is 0.227 at $l = 16$ (measured 0.226).
2. The across-input variance of $\log |h_l|^2$ saturates: $nV_l \rightarrow 44.6$, not $2 + 5l$.
3. The internal clock saturates at $t \approx 4/V$ once $\tau \gtrsim 2\sqrt{n}$, while the angle between inputs keeps decreasing.

4. The marginal skewness of deep pre-activations is dominated by the collective mode, $\gamma_1 \approx \frac{3}{2}\eta V_{l-1}r$, and the excess kurtosis is flat in r at $\approx 3\eta'V_{l-1}$, rather than following the one-body birth law, which contributes $O(n^{-1})$. Measured at width 1024: $\eta = 0.71\text{--}0.72$ at three depths, $\eta' = 0.57\text{--}0.63$, and $R^2 \approx 0.9$ for the linear law. Confirmed in form; the efficiency η is not yet derived.
5. The backward clock follows $\prod(1 - \theta_{k-1}/\pi)$, with exponent 3 in Fatou time.

8.4 Open problems

1. **Rates.** Prove Theorem 3.3(2) with explicit rates at fixed n , including the compression by the collective mode, and identify the law of the quenched clock fluctuations.
2. **The bent-crossing pair term at width n .** Compute its size and sign as a function of τ . It is the only genuinely two-body memory at third order, and it is the term stage 9's hub pairing did not contain.
3. **Radial–angular factorization beyond the input.** The input radius decouples exactly. How much of V_l (and hence of the collective memory) decouples at depth, and can the closure be run on the angular part with the radial mixture integrated exactly?
4. **A modular bound.** Is the closure error controlled by a modular quantity, for example the relative entropy of the layer state to its quasi-free projection (the negentropy of z_l), through a transport inequality on the chain strata?
5. **The radial efficiency.** Derive $\eta \approx 0.71$, the fraction of the across-input log-norm variance that acts as a common scale on the bulk noise, and its kurtosis analogue $\eta' \approx 0.6$. Both are constant across depths 8–16 at width 1024.
6. **Beyond the parabolic regime.** For $\tau \gg 2\sqrt{n}$ the radial temperature dominates. Does the two-tier dynamics there become the hyperbolic (Gauss-type) renormalization conjectured in stage 11?

Verification

The checks are run by `scripts/verify_stage12.py`; its output is saved next to this note. The network checks use one He network ($n = 1024$, $L = 16$, seed 5) and 2×10^5 standard inputs.

identity or prediction	result
1. Self-localization (Thm 3.3)	q_l, t_l , gate variance, V_l : see the table in Section 3; gate variance within 0.006 of $\theta_{l-1}/2\pi$ at all layers; $\Pr(r < 1) = 0.226$ against 0.227 at $l = 16$
2. Field lemma, clock recursion	$\mathbb{E}\Phi(r)\Phi(-r) = \arccos(\frac{t}{1+t})/2\pi$ to 10^{-8} at $t = 0.5, 2, 13$; $\mathbb{E}m_1^2/\mathbb{E}s_1^2$ equals the angle map to 10^{-6}
3. Temperature split (Prop. 3.7)	$l = 16$: $1 - q = 0.0779$ against $1 - e^{-V/4} \cos \theta_{16} = 0.0779$; $l = 4, 8, 12$ off by $+0.024, +0.010, -0.003$ (quenched scatter)
4. Fisher–Rao distance (Prop. 4.1)	numerical minimization equals $\sqrt{2} \operatorname{asinh}(r /\sqrt{2})$ to 10^{-6} at $r = 0.5, 4.29, -6.67$
5. Saturation of V (Prop. 4.2)	$(\frac{3}{2} - \frac{J_2}{2\pi})/\theta^2 = 0.99671, 0.99997$ at $\theta = 0.1, 0.01$; $nV_l = 30.76, 41.42, 44.60$ at $l = 16, 10^2, 10^5$; measured V_l at most 7% above prediction at every layer
6. Collective mode (Prop. 4.3)	trace share 0.008, 0.026, 0.058, 0.093 against $tV/4 = 0.005, 0.022, 0.053, 0.086$ at $l = 4, 8, 12, 16$
7. Collective-mode cumulant law	$\gamma_1 = 0.0252r, 0.0302r, 0.0336r$ at $l = 8, 12, 16$ ($R^2 = 0.86, 0.90, 0.91$; intercepts $\leq 10^{-4}$) against $\frac{3}{2}V_{l-1}r$: ratio $\eta = 0.72, 0.72, 0.71$. γ_2 flat in r ($R^2 \leq 0.11$) at 0.040, 0.053, 0.059 against $3V_{l-1}$ (ratio 0.57–0.63). Compression with η : $\operatorname{Var} r = 3.64, 7.11, 10.09$ against 3.79, 7.07, 10.14 measured
8. Ising couplings (Prop. 5.1)	log odds ratio 0.05614/0.05605, 0.03382/0.03388, 0.13388/0.13206 (exact/first order)
9. Wall-density law (Thm 5.3)	$\int \Phi(r)\Phi(-r)dr = 1/\sqrt{\pi}$ to 10^{-10} ; $\mathbb{E}\varphi = \sin(\Theta/2)/\sqrt{\pi}$ to 10^{-8} ; gate variance 0.022415 against 0.022396 at $t = 100$
10. Hermite cancellation (Prop. 5.5)	$\mathbb{E}\operatorname{He}_2\varphi = -(1+t)^{-3/2}/\sqrt{2\pi}$ to four digits at $t = 1, 13, 100$
11. Marginal memory (Section 5)	Gram–Charlier wall terms against the empirical closure error: correlation 0.983, 0.993, 0.996 at $l = 8, 12, 16$; residual $\leq 1.7 \times 10^{-4}$ per $ r $ -bin; error $< 5 \times 10^{-5}$ for $ r \geq 3$
12. Stratified third order (Thm 6.2)	softplus $\beta = 5$: 0.1242 vs 0.1224 (difference 0.0018 ± 0.0028); $\beta = 20$: 0.1677 vs 0.1698 (-0.0022 ± 0.0039)
13. Birth law (Prop. 6.4)	$c_3 = 0.326458$; $\int k_3^2 = 0.27002$; $\mathbb{E}k_3^2 = 0.02904$ against 0.02988 at $t = 13, 0.01512$ against 0.01523 at $t = 50$
14. Backward clock (Prop. 7.3)	measured 0.0108, 0.0800, 0.2489, 0.5571, 0.8798 against 0.0133, 0.0941, 0.2663, 0.5589, 0.8743 at $l = 0, 4, 8, 12, 15$; power law: 0.00131/0.00122 ($l = 10^2..10^3$), 0.00104/0.00102 ($l = 10^3..10^4$)

The folding angles and clocks: $\theta_l = 1.247, 1.054, 0.822, 0.584, 0.458, 0.378$ and $t_l^\infty = 0.47, 0.98, 2.13, 5.04, 8.72, 13.18$ at $l = 1, 2, 4, 8, 12, 16$; $\tau_{16} = \Phi_F(\theta_{16}) = 23.75$; $\tau_* = 2.0\sqrt{n} = 64$ at $n = 1024$.

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